Accessing Accelerometer Data on Nano 33 IoT

Learn how to measure the relative position of the Nano 33 IoT through the LSM6DS3 IMU sensor.

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IMU Module

This tutorial will focus on the 3-axis accelerometer sensor of the **LSM6DS3** module on the Arduino Nano 33 IoT, in order to measure the relative position of the board. This will be achieved by utilizing the values of the accelerometer's axes and later print the return values through the Arduino IDE Serial Monitor.

Goals

The goals of this project are:

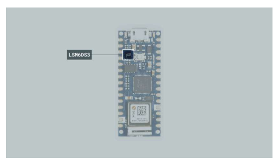
- Understand what an LSM6DS3 module is.
- Use the LSM6DS3 library.
- Read the raw data of the accelerometer sensor.
- Convert the raw data into board positions.
- Print out live data through the Serial Monitor.

Hardware & Software Needed

- This project uses no external sensors or components.
- In this tutorial we will use the Arduino Create Cloud Editor to program the board.

The LSM6DS3

IMU stands for: inertial measurement unit. It is an electronic device that measures and reports a body's specific force, angular rate and the orientation of the body, using a combination of accelerometers, gyroscopes, and oftentimes magnetometers. In this tutorial we will learn about the LSM6DS3 IMU module, which is included in the Arduino Nano 33 IoT Board.



The LSM6DS3 sensor.

The LSM6DS3 is a system-in-package featuring a 3D digital linear acceleration sensor and a 3D digital angular rate sensor.

The LSM6DS3 Library

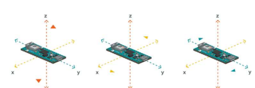
The Arduino LSM6DS3 library allows us to use the Arduino Nano 33 IoT IMU module without having to go into complicated programming. The library takes care of the sensor initialization and sets its values as follows:

- **Accelerometer** range is set at -4 |+4 g with -/+0.122 mg resolution.
- **Gyroscope** range is set at -2000 | +2000 dps with +/-70 mdps resolution.
- **Output** data rate is fixed at 104 Hz.

If you want to read more about the LSM6DS3 sensor module see **here**.

Accelerometer

An accelerometer is an electromechanical device used to measure acceleration forces. Such forces may be static, like the continuous force of gravity or, as is the case with many mobile devices, dynamic to sense movement or vibrations.



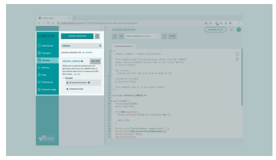
How the accelerometer works.

In this example, we will use the accelerometer as a "level" that will provide information about the position of the board. With this application we will be able to read what the relative position of the board is, as well as the degrees by tilting the board up, down, left or right.

Creating the Program

1. Setting up

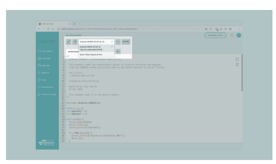
Let's start by opening the Arduino Cloud Editor, click on the **Libraries** tab and search for the **LSM6DS3** library. Then in **>** **Examples**, open the and once it opens, rename it as **Accelerometer**.



Finding the library in the Cloud Editor.

2. Connecting the board

Now, connect the Arduino Nano 33 IoT to the computer and make sure that the Cloud Editor recognizes it, if so, the board and port should appear as shown in the image below. If they don't appear, follow the instructions to install the plugin that will allow the Editor to recognize your board.



Selecting the board.

3. Printing the relative position

Now we will need to modify the code on the example, in order to print the relative position of the board as we move it in different angles.

Let's start by initializing the the x, y, z axes as `float` data types, and the `int degreesX = 0;` and `int degreesY = 0;` variables before the `setup()`.

In the `setup()` we should **remove** the following lines of code:

```
Serial.println();
Serial.println("Accele
Serial.println("X\tY\t
```

Since the raw values of the three axes will not be required, we can remove the lines which will print these. Similarly, we should **remove** the following lines from the `loop()`:

```
Serial.print(x);
Serial.print('\t');
Serial.print(y);
Serial.print('\t');
Serial.println(z);
```

Instead, in the `loop()` we tell the sensor to begin reading the values for the three axes. In this example we will not be using the readings from the Z axis as it is not required for this application to function, therefore you could remove it.

After the `IMU.readAcceleration` initialization, we add four `if` statements for the board's different positions. The statements will calculate the direction in which the board will be tilting towards, as well as provide the axe's degree values.

```
if(x > 0.1){
    x = 100*x;
    degreesX = map(x
    Serial.print("T:
    Serial.print(deg
    Serial.println('
    }
  if(x < -0.1){
    x = 100*x;
    degreesX = map(x
    Serial.print("T:
    Serial.print(deg
    Serial.println('
    }
  if(y > 0.1){
    y = 100*y;
    degreesY = map(y
    Serial.print("T:
    Serial.print(deg
    Serial.println('
    }
  if(y < -0.1){
    y = 100*y;
    degreesY = map(y
    Serial.print("T:
    Serial.print(deg
    Serial.println('
    }
```

Lastly, we `Serial.print` the results value and add a `delay(1000);`.

> *Note: For the following code to properly work, the board's facing direction and inclination during the initialization of the code, need to be specific. More information will be shared on the "testing it out" section.*

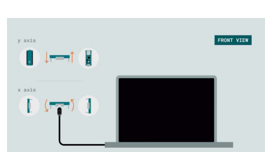
4. Complete code

If you choose to skip the code building section, the complete code can be found below:

```
/*
  Arduino LSM6DS3 -

  This example read
  from the LSM6DS3 s

  The circuit:
  - Arduino Nano 33

  Created by Riccar

  Modified by Jose (
  27 Nov 2020

  This example code
*/

#include <Arduino_L

float x, y, z;
int degreesX = 0;
int degreesY = 0;

void setup() {
  Serial.begin(9600
  while (!Serial);
  Serial.println("S
```

Testing It Out

In order to get a correct reading of the board data, before uploading the sketch to the board hold the board in your hand, from the side of the USB port. The board should be facing up and "pointing" away from you. The image below illustrates the board's position and how it works:

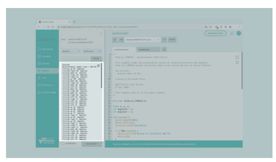


Interacting with the X and Y axes.

Now, you can verify and upload the sketch to the board and open the Monitor from the menu on the left.

If you tilt the board upwards, downwards, right or left, you will see the results printing every second according to the direction of your movement!

Here is a screenshot of the sketch returning these values:



Printing out the "tilt condition" of the board.

Troubleshoot

Sometimes errors occur, if the code is not working there are some common issues we can troubleshoot:

- Missing a bracket or a semicolon.
- Arduino board connected to the wrong port.
- Accidental interruption of cable connection.
- The initial position of the board is not as instructed. In this case you can refresh the page and try

Conclusion

In this simple tutorial we learned what an IMU sensor module is, how to use the **LSM6DS3** library, and how to use an Arduino Nano 33 IoT to get data. Furthermore, we utilized the 3-axis accelerometer sensor, in order to measure and print out the degrees and relative position of the board.

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